Hypergraph-based modeling of virtual systems

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Description of simulation (robotic arm)

```
[anthropomorphic_arm  import meta_kinematics.himeko"]
robot use kinematics.elements use kinematics.geometry use kinematics.axes
{
    link_color [0.7, 0.2, 0.4, 1.0]
    base_link: kinematics.elements.link {
        mass 25.0
        link_geometry: cylinder {
            dimension [0.13, 0.05]
        }
        visual -> link_geometry
        collision -> link_geometry
        color -> robot.link_color
        origin [0.0, 0.0, 0.025]
    }
    link_0: kinematics.elements.link {
        mass 5.0
        link_geometry: cylinder {
            dimension [0.075, 0.2]
        }
...
    @jtool: kinematics.rev_joint  {
        <- link_4, [0.0, 0.0, 0.15] -> tool,
        -> AXIS_Z
    }
}
```

Description of virtual space (factory setting)

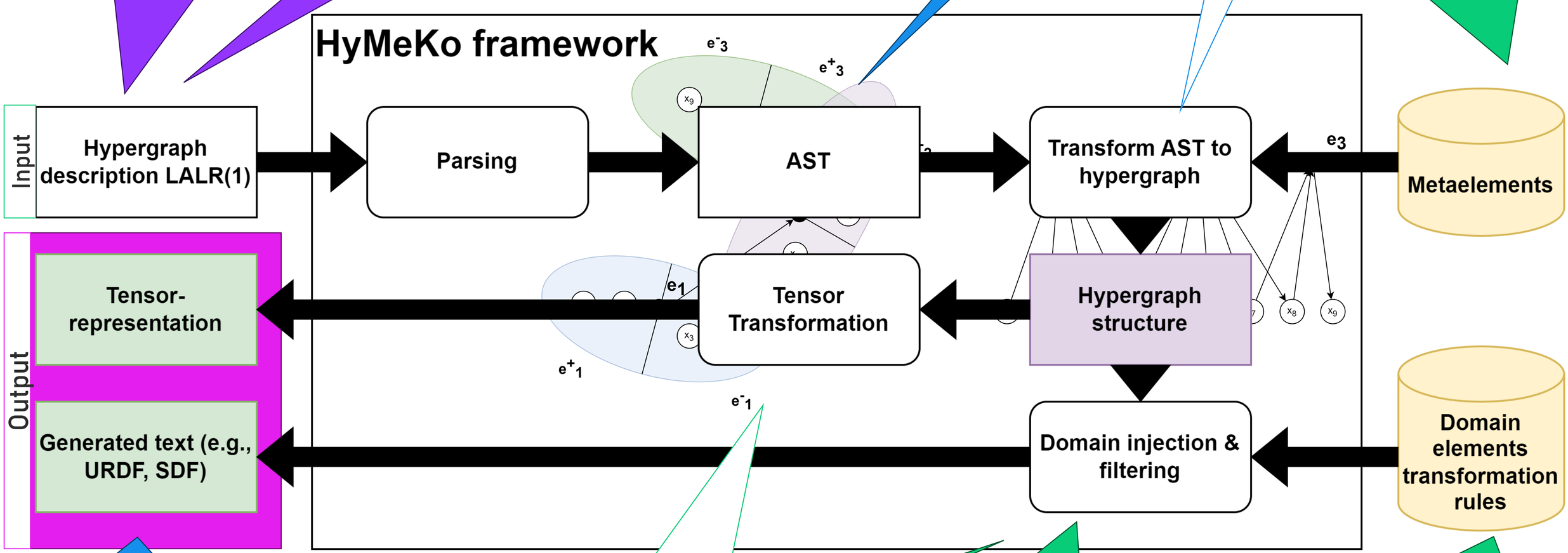
```
[mxw_cylinder import "mxw_meta.himeko"]
munkahenger_scene use maxwhere.elements {
    units -> maxwhere.units.design
    munkahenger: -> mxw_node {
        position [600, 0, 600]
        orientation [90, 0, 0]
        scale [10, 10, 10]
        munkahenger_base: -> mxw_node {
            mesh {
                url "munkahenger_base.mesh"
            }
        }
        munkahenger_moving: -> mxw_node {
            position [0, 0, 20]
            mesh {
                url "munkahenger_moving.mesh"
            }
        }
    }
}
```

More concise & compact description than XML and JAML

Abstract syntax-tree (AST) representing tree-based description of hypergraph elements

(Directed)Hypergraph form processable by Python

Other hypergraph model describing elements of a specific domain injected for transformation



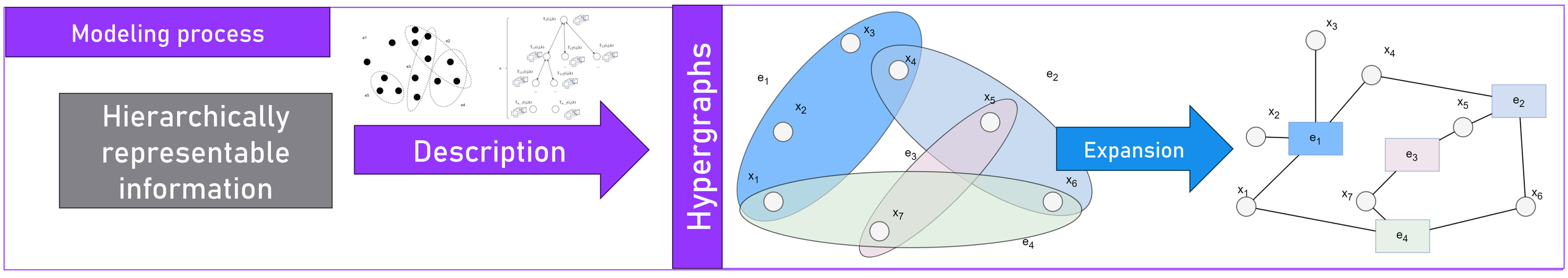
Text-based output for simulators (e.g., MaxWhere, Gazebo), explicitly mapped from the hypergraph-description

Tensor-representation of the hypergraph

Visiting graph elements, create a text output for every element

Using the domain-specific rules, transform hypergraph to another graph

Programmatic rules on visiting and matching hypergraph elements of certain properties (e.g., *stereotype of*)



References

- [[1] A. Csapo, A. Sudár, B. Berki, B. Gergely, B. Berenyi, and C. Czaban,"Measuring Virtual Rotation Skills in MaxWhere," Sept. 2020.
- [2] D. Tola and P. Corke, "Understanding URDF: A Survey Based on UserExperience," 2023. _eprint: 2302.13442.
- [3] N. Albergo, V. Rathi, and J.-P. Ore, "Understanding Xacro Misunderstandings,"ArXiv, vol. abs/2109.09694, 2021.
- [4] A. Bretto, Hypergraph Theory: An Introduction. Springer Publishing Company,Incorporated, 2013

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